

# Yazan Alatrach

## Experience

### **Analytics Engineer - Octopus Energy**

Sep '24 - Present

- Owned transformation layer architecture using dbt and SQL to build scalable data models supporting financial reporting, energy trading operations, and risk management across international regions, processing millions of daily events from transactional databases, API logs, and operational systems.
- Established technical standards and best practices for data modeling, development, testing, and monitoring, implementing robust change management, source control, and code review processes to ensure data quality and cent-level precision for finance teams.
- Refactored obscure legacy code into a transparent, maintainable system using Python and dbt, reducing processing time by 80% and improving overall system reliability for business-critical energy market operations.
- Enabled cross-functional stakeholders by collaborating directly with finance, operations, and risk management teams to understand requirements and deliver reliable data products via BI tools and operational dashboards, supporting real-time decision-making.
- Designed and implemented scalable ELT pipelines using Python, Apache Airflow, and DBT on AWS infrastructure (S3, EKS/Kubernetes), creating repeatable integration patterns with BI tools and reverse ETL processes for consistent metric reporting.
- Built comprehensive monitoring and alerting systems for data pipelines to quickly identify failures and data anomalies, ensuring stability and data quality across the transformation layer.
- Engineered integration between energy storage infrastructure and financial reporting systems, developing scripts to allocate battery capacity and accurately reflect impact in financial reporting.

### **Senior Data Engineer - Capital One**

Aug '21 - May '23

- Built and maintained transformation layer data models using SQL and Snowflake to support diverse business needs, from daily operational metrics to historical trend analysis, enabling stakeholders across finance and operations to access critical insights.
- Optimized critical business queries, reducing query execution time by 50% from 1 hour to 30 minutes, enabling swifter access to key metrics like application submissions, conversion, and fund rates, thereby accelerating daily and historical data analysis.
- Designed and evolved data models to accommodate new business requirements and data sources, demonstrating ability to work with ambiguity and adapt data structures for evolving analytical needs.
- Developed robust ETL/ELT pipelines using Snowflake and Apache Airflow to integrate data from CRMs, payment systems, and transactional databases, establishing repeatable patterns for data ingestion and transformation.
- Collaborated directly with data analysts and business users to understand requirements, create and refine Tableau dashboards, and deliver actionable data products that elevated reporting quality and supported informed decision-making.

### **“Network”/Software Engineer - Cisco Systems**

Jul '19 - Aug '21

- Created a Python and Jenkins-based testing suite, automating feature verification and deployment, cutting testing duration from 15 hours to just 2 hours.
- Automated interoperability tests between Voice over IP (VoIP) phones and Webex clients, confirming reliable calling features, enhancing user experience.
- Led training sessions to share test results and methodologies with team members, promoting collaborative learning and procedural efficiency.
- Engaged with both domestic and international clients to resolve Webex client issues, handling feature configurations and ensuring customer satisfaction.
- Developed and deployed Webex clients to support increased demand for remote work during the COVID-19 pandemic, creating a bash script for efficient IP address monitoring across server regions in the United States, resulting in significant time savings for the team.
- Conducted manual testing of (VoIP) phones to ensure feature compatibility with various software updates, utilizing Wireshark for network trace analysis.

## Education

Georgia Institute of Technology  
The University of Texas at Austin

**M.S** Computer Science  
**B.S** Electrical/Computer Engineering

## Skills and Certifications

Ex-AWS Certified Solutions Architect Associate

Languages: SQL, Python, Unix/Linux, Javascript, Swift

Tools/Frameworks: DBT, AWS, Airflow, Databricks, Streamlit, Snowflake, Jenkins, GCP, Tableau, FastAPI, Jenkins